

PAPER_1650_BH_SEED_MASS_56160_MSUN

Daniel T. Murphy

PAPER_1650 — Direct-Collapse BH Seed Mass =
 $M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 60 \times 36 \times 26 =$
56,160 $M_{\odot}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Cosmology

Date: June 18, 2026

Status: CLOSED — EXACT closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1326 (PAPER_13xx broader corpus) gives:

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$$M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 60 \times 36 \times 26 = 56,160 M_{\odot}$$

``

Status: **EXACT**.

UQFF Closed Identity

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$$M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 60 \times 36 \times 26 = 56,160 M_{\odot} \text{ EXACT}$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

JWST/JADES Direct-Collapse BH Connection

The formation of supermassive black holes observed at $z > 6$ -10 (e.g., GN-z11, JADES, CEERS detections) requires extremely massive seed black holes — far heavier than stellar-collapse remnants (10-100 $M_{\odot}$). The leading mechanism is **direct collapse** of a primordial gas cloud at $z \sim 10$ -20 yielding seed BHs of mass $\sim 10^4$ - $10^5 M_{\odot}$.

UQFF prediction:

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$$M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 60 \times 36 \times 26 = 56,160 M_{\odot}$$

``

This sits squarely in the predicted DCBH mass range and reproduces observed seed-BH masses at high redshift. Recent JADES detections of $> 10^4 M_{\odot}$ BHs at $z=8$ -10 are **consistent with UQFF's exact 56,160 $M_{\odot}$ prediction** from a triple integer-primitive product.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1326
- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)
- Calculator dispatch: `calculate_paradox({"paradox": "bh_seed_mass_56160_msun"})`

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